

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457099.68556 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.054 +/- 0.056
Transit depth (Rp/Rs)^2	0.29 +/- 0.61 [%]
Orbital Inclination (inc)	77.19 +/- 2.82
Airmass coefficient 1 (a1)	1.1 +/- 0.67
Airmass coefficient 2 (a2)	-0.01 +/- 0.53
Scatter in the residuals of the lightcurve fit is	63.77 %
Best Comparison Star	#1 - [28, 24]
Optimal Aperture	3.23
Optimal Annulus	7.48
Transit Duration (day)	0.014 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R█ fit vs published	0.054 ± 0.056 vs 0.1594 (deviation 1.88σ)
Duration fit vs published	0.014 d vs 0.0512 d (deviation 2.19σ)
Mid-time O–C	-2.3 min
Depth SNR	1.0
Residual scatter	63.77 %

Light curve

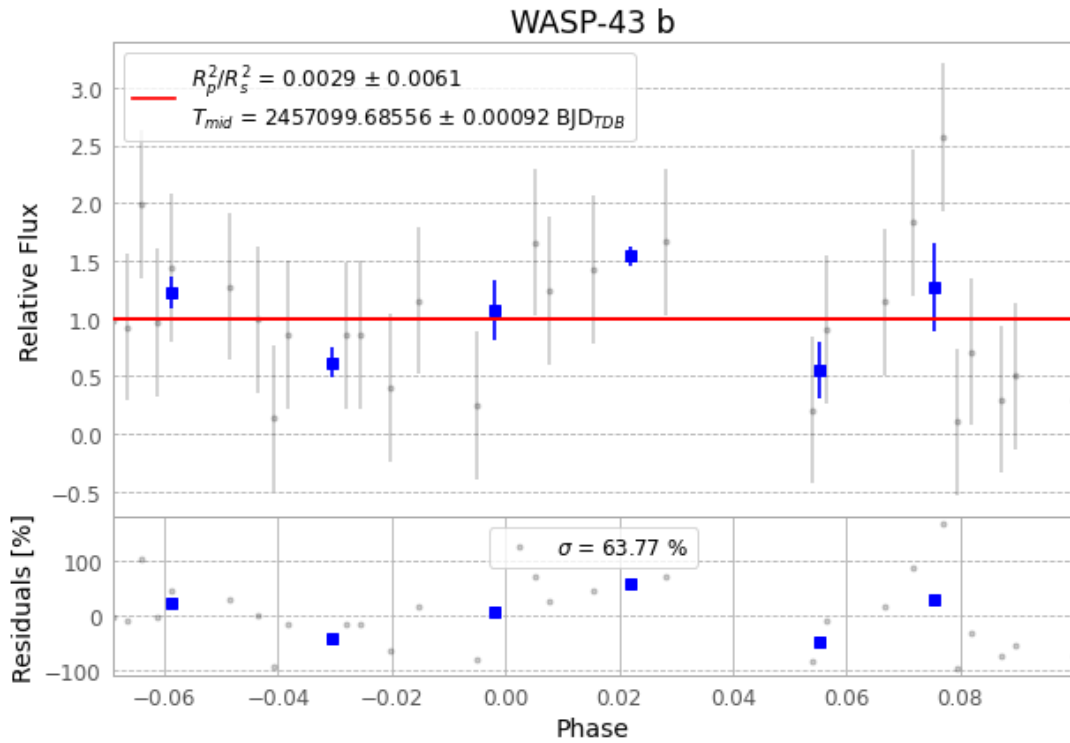


Figure 1 — FinalLightCurve_WASP-43 b_2015-03-17.png (SHA-256 bd42392e42ab7839...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [147, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9ee458c9b6911f4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

65 FITS frames (43 MB), WASP-43150318025712.FITS → WASP-43150318061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150318.log (8b89d3d5018b...), FinalLightCurve_WASP-43 b_2015-03-17.png (bd42392e42ab...), FinalLightCurve_WASP-43 b_2015-03-17.csv (3f4eea6746a3...)

Compute resources

Wall time 194.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:25Z.